

Applying the Business Model Canvas to the Retail Sector: Opportunities and Challenges of Augmented Reality

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Abstract

The paper follows a profound research effort about the adoption of Augmented Reality in the Retail sector. To better understand the potentialities and weaknesses of such technology, Business Model Canvas was applied to the case of a retailer aspiring to introduce AR in its business. The analysis highlights that AR holds a prosper future, with abilities in increasing sales and reducing return rates by enhancing customer experience. At the same time, results expose important challenges such as implementation costs, technological barriers and the need for partnerships with software and hardware companies. Overall, the findings suggest a prosper future for AR in Retail sector, but its success will depend on the abilities of retailers to balance customer value creation with sustainable business.

Introduction

In the following paragraphs we will investigate the theoretical framework of Business Model Canvas, the industrial sector of Retail and Augmented Reality (AR) and consequently develop our application of this theoretical tool to the current framework.

Business Model Canvas

Markets in our complex society seem to be complex systems. If one wants to “succeed” with his/her business of any kind, not all routes are successful. It is important to be fast and precise in decision taking. Patterns appear to emerge, there is an “adaptive” logic that businesses can follow to succeed.

To do so, it is necessary to expand the personal knowledge through primary and secondary research, market analysis, customer analysis and macro analysis.

Business Model Canva (BMC) appears to emerge as one of the most balanced tools to help in this job. BMC defines nine building blocks first identified by Alexander Osterwalder in his PhD thesis in 2004 (Osterwalder, 2004):

1. Value Proposition: the set of products and services provided solving customer needs.
2. Customer Segments: the groups of people or businesses that are served.
3. Customer Relationships: the type of interaction with the clients and its value.
4. Channels: ways to deliver value and communicate with clients.
5. Key Resources: essential assets needed to support the business.
6. Key Activities: core activities to create value.
7. Key Partnerships: external parties providing resources or activities.
8. Cost Structure: all costs to operate.

9. Revenue Streams: ways of getting incomes.

Retail and Augmented Reality

Following technological innovations, layers of reality have emerged, there is a shade starting from the Real Environment and getting to Virtual Reality. Augmented Reality (AR) is physical reality in which participants also see virtual elements. (Arena F, 2022)

The mechanism of AR is simple, hardware components detect real world information and show the user the environment, with some details added through software.

Hardware includes sensors such as cameras, accelerometers, GPS, and depth sensors, in combination with displaying devices like smartphones, tablets, or head-mounted glasses. Software process sensor data, handle image recognition, object tracking, and overlay virtual elements accurately. This can potentially lead to various innovations in medicine applications, education, tourism, gaming and entertainment. However, AR faces limitations like hardware power supply, while software may struggle with cross platform compatibility.

Application of the theoretical framework to the industrial sector

To apply the theoretical framework to Retail sector, we consider a general retail shop who would like to introduce Augmented Reality technologies to increase the number of sales. In the following we delve deeply into the nine building blocks.

Value Proposition: The main value is brought by the immersion of the consumer in such activity; this increases curiosity and entertainment with the shopping activity (online and offline), increasing satisfaction and brand loyalty (Joy Onma Enyejo, 2024).

The second main aspect is information provision; the possibility to add real-time pop-ups shown on the physical products in store, or

the ability of building a guided experience increase the amount of information provided to the customer, solving clients doubts about the products instantly and increasing product perceived value. (Joy Onma Enyejo, 2024)

Customer Segments: Most interesting clients are young demographics which are tech-savvy, but in the future the segment is expected to widen with time and diffusion of AR. To support such argument, one can notice that big tech companies, like Snapchat are, investing in AR features, saying that most Gen-Z people would like to use some during shopping activities. (Walk-Morris, 2022)

Customer Relationships: One must mark the facilitation of online purchases as a possible negative impacting factor, leading to a not very intensive customer relationship. However, additional information and increased customer confidence can foster brand loyalty, leading to higher return to the shop/website rates (Joy Onma Enyejo, 2024). Overall, the relationship relies on ongoing digital interaction, using push notifications and in-app support to assist users.

Channels: Users can access their smartphones and tablets and enter mobile apps or WebAR platforms. For example, IKEA Place is an iOS/Android app built on ARKit (IKEA, 2017). However, social media (as Snapchat and Instagram) can play an important role in fastening the connection between the online platforms and clients through interactive advertisements. (Snapchat)

Moreover, through the collection of data the client experience can be personalized, increasing engagement and satisfaction.

Key Resources: Of fundamental importance are software and hardware components. In first place, app or web platform software, 3D models, graphics, customer data. Secondly, tablets or AR glasses are fundamental if the desire is to implement AR in-store experience, while for online shopping

the device is already in the hands of the client; however this requires AR models to be runnable on all possible devices to reach as many people as possible. Human resources are also necessary, such as technicians, software developers and user-interface designers.

Key Activities: money and effort must be invested to create new software, implement and test them. Moreover, companies should be able to fix and document problems for proper functioning. Software must be entertaining and information provided must be clear, fundamental to engage clients and foster their confidence in buying the product.

Key Partnerships: from Key Resources and Key activities one can notice the need of interacting with hardware companies, software companies (Android, Apple), social media, technicians and user-interface designers.

Cost Structure: Hardware need to be bought, software to be created or to be paid from monthly subscriptions (an example is XR Brand), models and contents must be personalized for each shop and/or customer, electric current in case of in-site hardware. These costs are R&D and fixed costs.

Moreover, marketing costs for advertisements.

Revenue Streams: The reduced uncertainty during buying process implies a reduced return rate (Hou, 2024), offering a realistic try-before-you-buy experience and increasing customer confidence.

Moreover, according to a report by PwC (2023), retailers that offer virtual try-on features saw a 30% reduction in product returns and a 40% increase in conversion rates. (Celestin, 2024)

Conclusion

AR implementation seems to be feasible and rewarding, leading to high customer satisfaction levels and higher incomes. It is a feature already spreading into different sectors, it will lead to a big innovation: once customers get used to shops using AR, they will start to

expect from every shop to provide such experience, and most retailers will have to

adapt to this big change which will probably become a new standard.

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ChatGPT was used for fastening and improving the research of data on the web, it was used only as a side tool, not producing any line of text or any content.